THE BASE CHANGE MAPS OF TWO SIDE BY SIDE CARTESIAN SQUARES COMPOSE TO THE BASE CHANGE MAP OF THE OUTER RECTANGLE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0ATR (sources/stacks-duality/pinned-original-source.tex line 1375).

Allowed standard prerequisites (editorial).

- The base change map of a cartesian Tor independent square. Same chapter, equation labelled equation-base-change-map and Lemma labelled lemma-twisted-inverse-image, cited at proof lines 1395-1404. Supplies the three gamma maps whose composition is asserted.
- The dual statement for ordinary base change maps in the horizontal direction. Cohomology, Remark labelled cohomology-remark-compose-base-change-horizontal, cited at proof lines 1464-1466. Identifies the final reduced arrow with xi_rect.
- Coherence of compositions of functors in a two category. Categories, Lemma labelled categories-lemma-properties-2-cat-cats and Definition labelled categories-definition-horizontal-composition, cited at proof lines 1450-1454. Gives the commutativity of the quadrilaterals in the chase.

Source reference. Original equation 0AA6: sources/stacks-duality/pinned-original-source.tex, source lines 1192–1192. Same-chapter pointer into the author environment equation at source line 1192; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0A9E: sources/stacks-duality/pinned-original-source.tex, source lines 653–653. Same-chapter pointer into the author environment lemma at source line 653; invoked at proof line 1396 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 003F: sources/stacks-chow/tags.txt, source lines 132–132. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1456 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original definition 003E: sources/stacks-chow/tags.txt, source lines 131–131. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1458 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0ATM: sources/stacks-chow/tags.txt, source lines 13620–13620. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 1466 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Consider a commutative diagram*

$$\begin{array}{ccccc} X'' & \xrightarrow[g']{} & X' & \xrightarrow{g} & X \\ {\scriptstyle f''}\downarrow & & {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y'' & \xrightarrow{h'} & Y' & \xrightarrow{h} & Y \end{array}$$

of quasi-compact and quasi-separated schemes where both diagrams are cartesian and where f and h as well as f' and h' are Tor independent. Then the maps (0AA6) for the two squares compose to give the base change map for the outer rectangle (see proof for a precise statement).

Proof. It follows from the assumptions that f and $h \circ h'$ are Tor independent (details omitted), hence the statement makes sense. In this proof we write g^* in place of Lg^* and f_* instead of Rf_*. Let a, a', and a'' be the right adjoints of Lemma 0A9E for f, f', and f''. The arrow corresponding to the right square is the composition

$$\gamma_{right} : g^* \circ a \to g^* \circ a \circ h_* \circ h^* \xleftarrow{\xi_{right}} g^* \circ g_* \circ a' \circ h^* \to a' \circ h^*$$

where $\xi_{right} : g_* \circ a' \to a \circ h_*$ is an isomorphism (hence can be inverted) and is the arrow "dual" to the base change map $h^* \circ f_* \to f'_* \circ g^*$. The outer arrows come from the canonical maps $1 \to h_* \circ h^*$ and $g^* \circ g_* \to 1$. Similarly for the left square we have

$$\gamma_{left} : (g')^* \circ a' \to (g')^* \circ a' \circ (h')_* \circ (h')^* \xleftarrow{\xi_{left}} (g')^* \circ (g')_* \circ a'' \circ (h')^* \to a'' \circ (h')^*$$

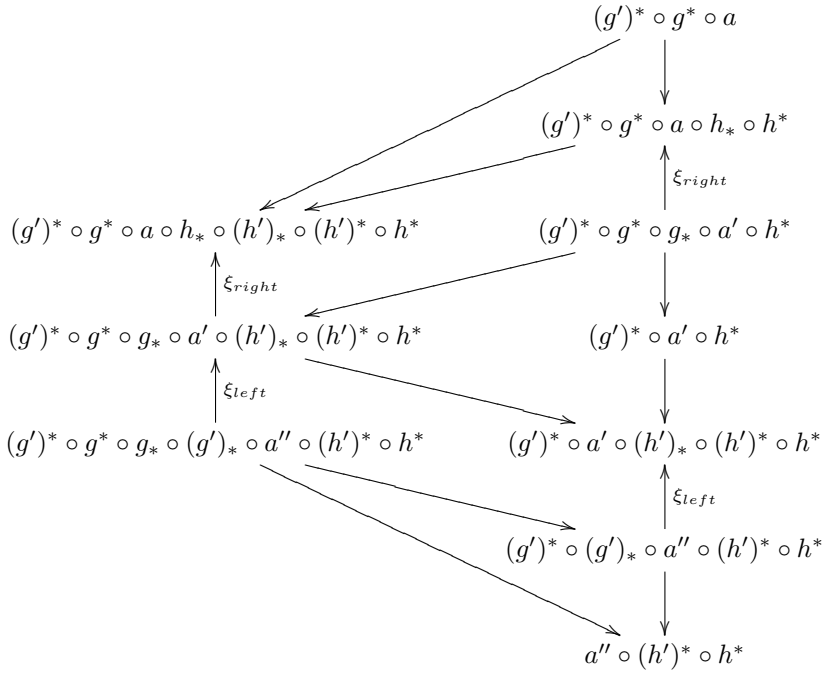
For the outer rectangle we get

$$\gamma_{rect} : k^* \circ a \to k^* \circ a \circ m_* \circ m^* \xleftarrow{\xi_{rect}} k^* \circ k_* \circ a'' \circ m^* \to a'' \circ m^*$$

where $k = g \circ g'$ and $m = h \circ h'$. We have $k^* = (g')^* \circ g^*$ and $m^* = (h')^* \circ h^*$. The statement of the lemma is that γ_{rect} is equal to the composition

$$k^* \circ a = (g')^* \circ g^* \circ a \xrightarrow{\gamma_{right}} (g')^* \circ a' \circ h^* \xrightarrow{\gamma_{left}} a'' \circ (h')^* \circ h^* = a'' \circ m^*$$

To see this we contemplate the following diagram



Going down the right hand side we have the composition and going down the left hand side we have γ_{rect}. All the quadrilaterals on the right hand side of this diagram commute by Categories, Lemma 003F or more simply the discussion preceding Categories, Definition 003E. Hence we see that it suffices to show that

$$g_* \circ (g')_* \circ a'' \xrightarrow{\xi_{left}} g_* \circ a' \circ (h')_* \xrightarrow{\xi_{right}} a \circ h_* \circ (h')_*$$

is equal to ξ_{rect}. This is the statement dual to Cohomology, Remark 0ATM and the proof is complete. □